

Enhancing BI-RADS Classification in Digital Mammography

A Vision-Language Foundation Model Approach using
Mammo-CLIP on the VinDr-Mammo Dataset

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Presentation Outline



Introduction &
Background

Problem Statement

Limitations of
Traditional Models

Proposed Solution:
Mammo-CLIP

Dataset Overview

Core Methodology

Experimental Setup

Expected Results

Clinical Impact

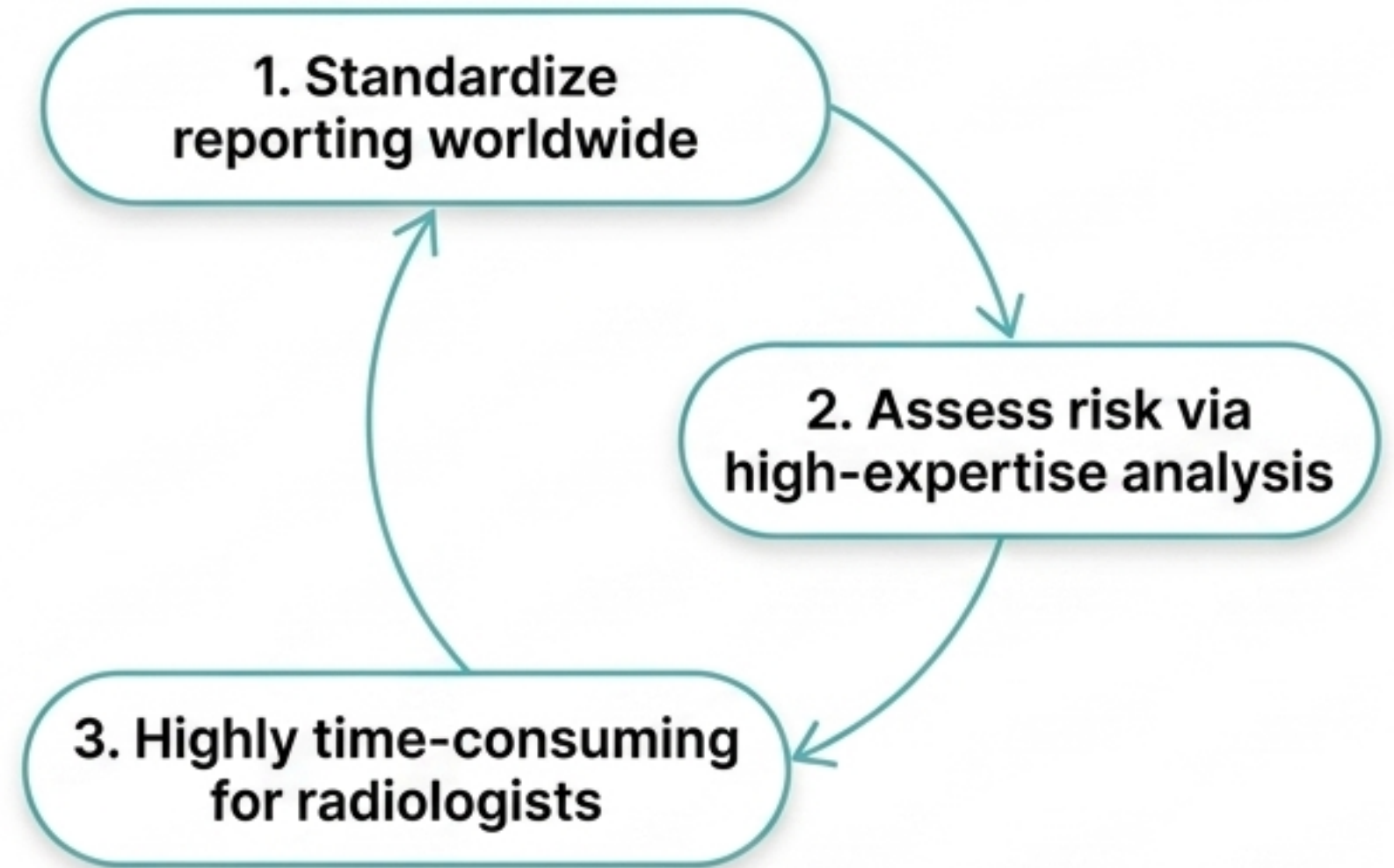
Conclusion & Future
Work

Background: The Role of Mammography

Breast Cancer: The most prevalent malignancy among women globally.

Early Detection: Screening mammography is the gold standard for reducing mortality rates.

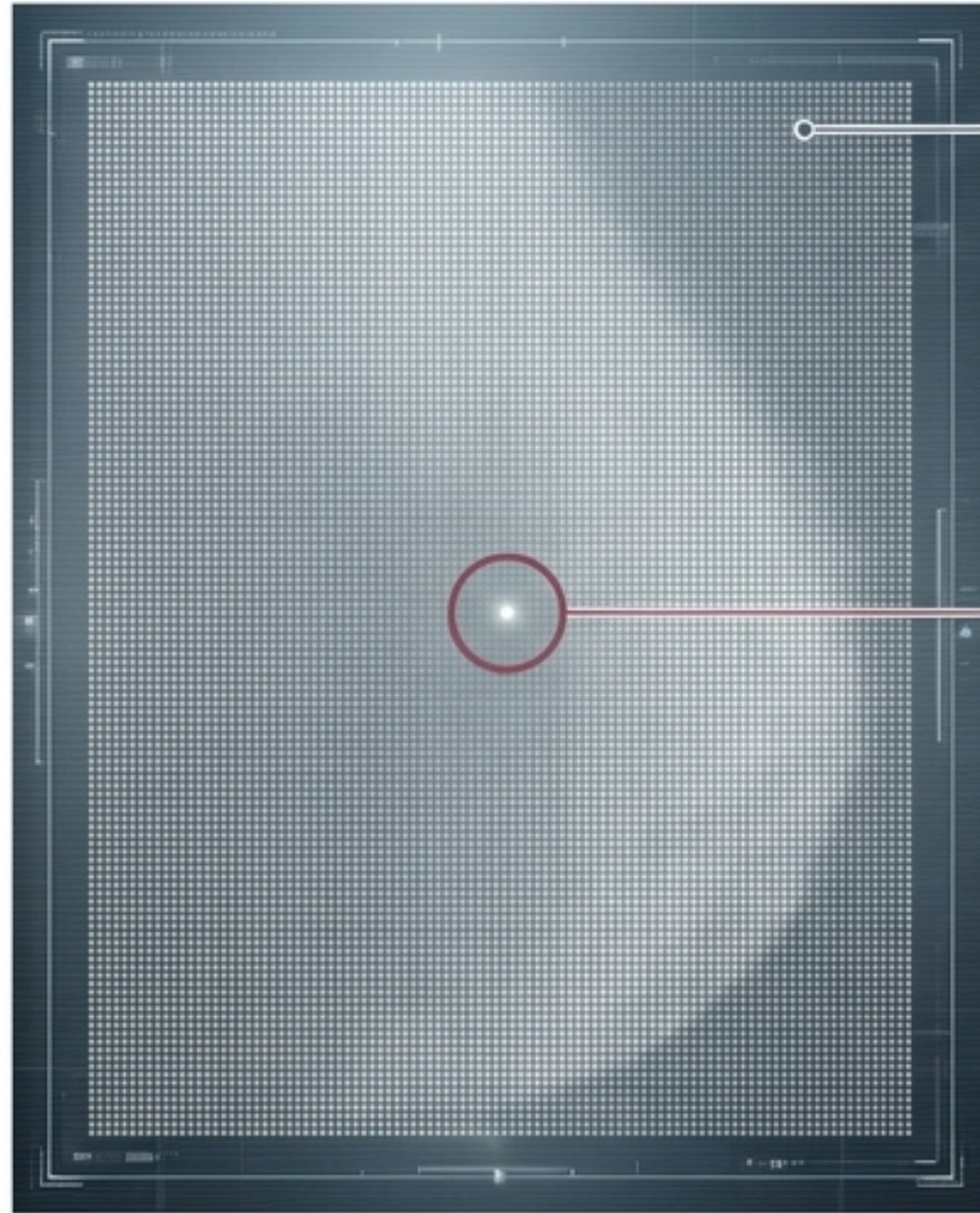
The BI-RADS System



The Challenge: Interpreting mammograms requires high expertise and is highly time-consuming for radiologists.

Challenges in Mammogram Interpretation

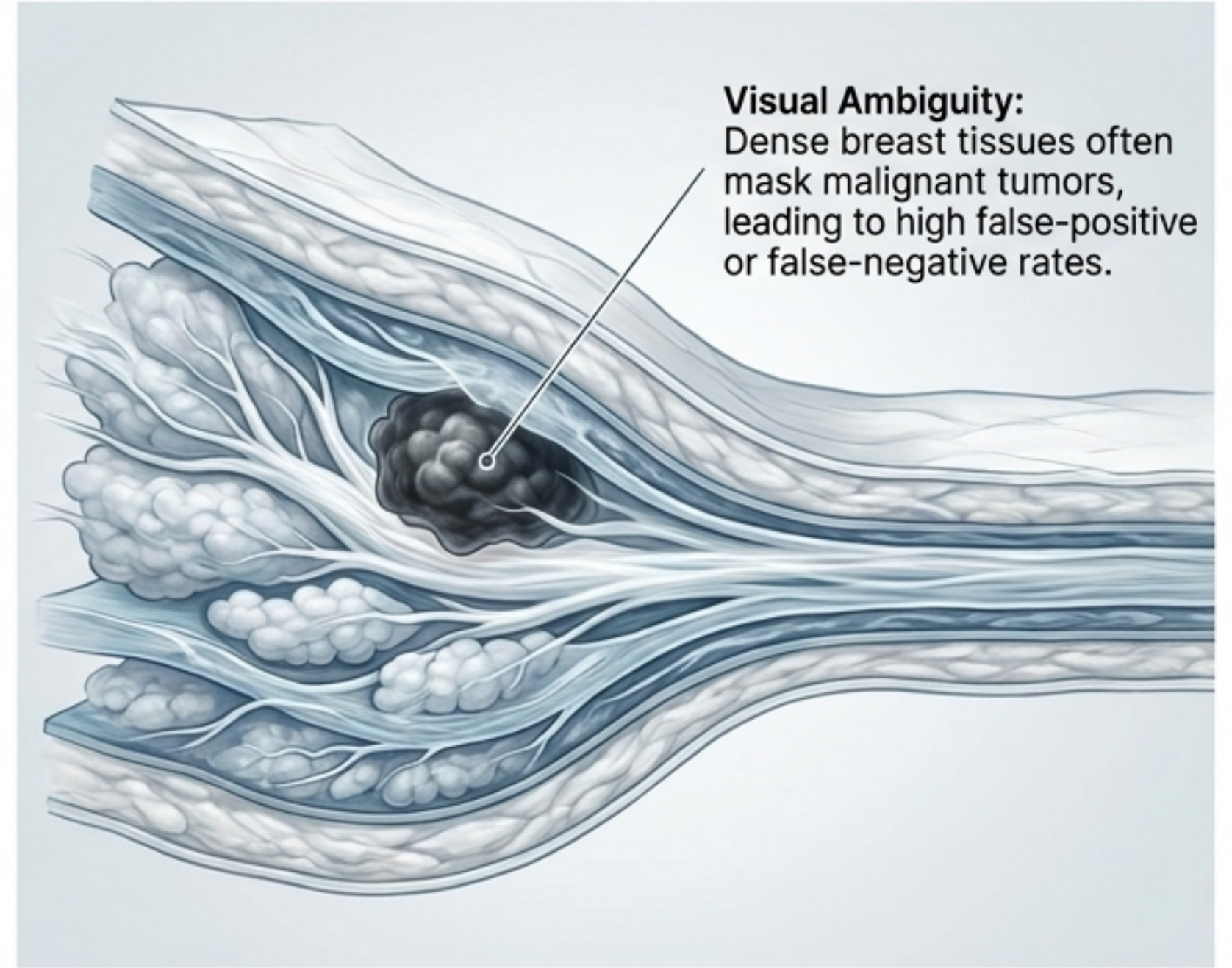
High-Resolution Complexity & Subtle Lesions



High-Resolution Complexity:
Mammograms are massive images (often > 3000x4000 pixels).

Subtle Lesions:
Early signs like microcalcifications are extremely small and easily missed.

Visual Ambiguity



Visual Ambiguity:
Dense breast tissues often mask malignant tumors, leading to high false-positive or false-negative rates.

Need for AI: Computer-Aided Diagnosis (CADx) systems are urgently needed to act as a reliable second reader.

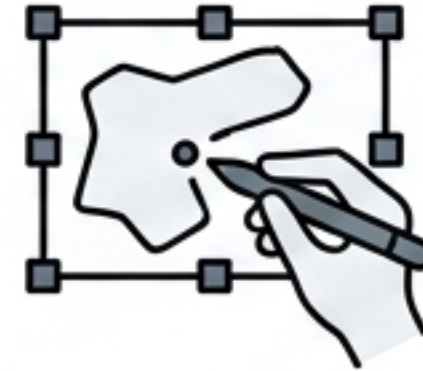
Why Traditional CNNs Fall Short



High-Resolution Bottleneck



Standard CNNs heavily downsample images (e.g., to 224×224), causing critical loss of diagnostic details.



Annotation Burden



CNNs require massive amounts of expensive, pixel-level annotations (bounding boxes/masks) by medical experts.



Locality Bias



CNNs struggle to connect global structural context across the breast.

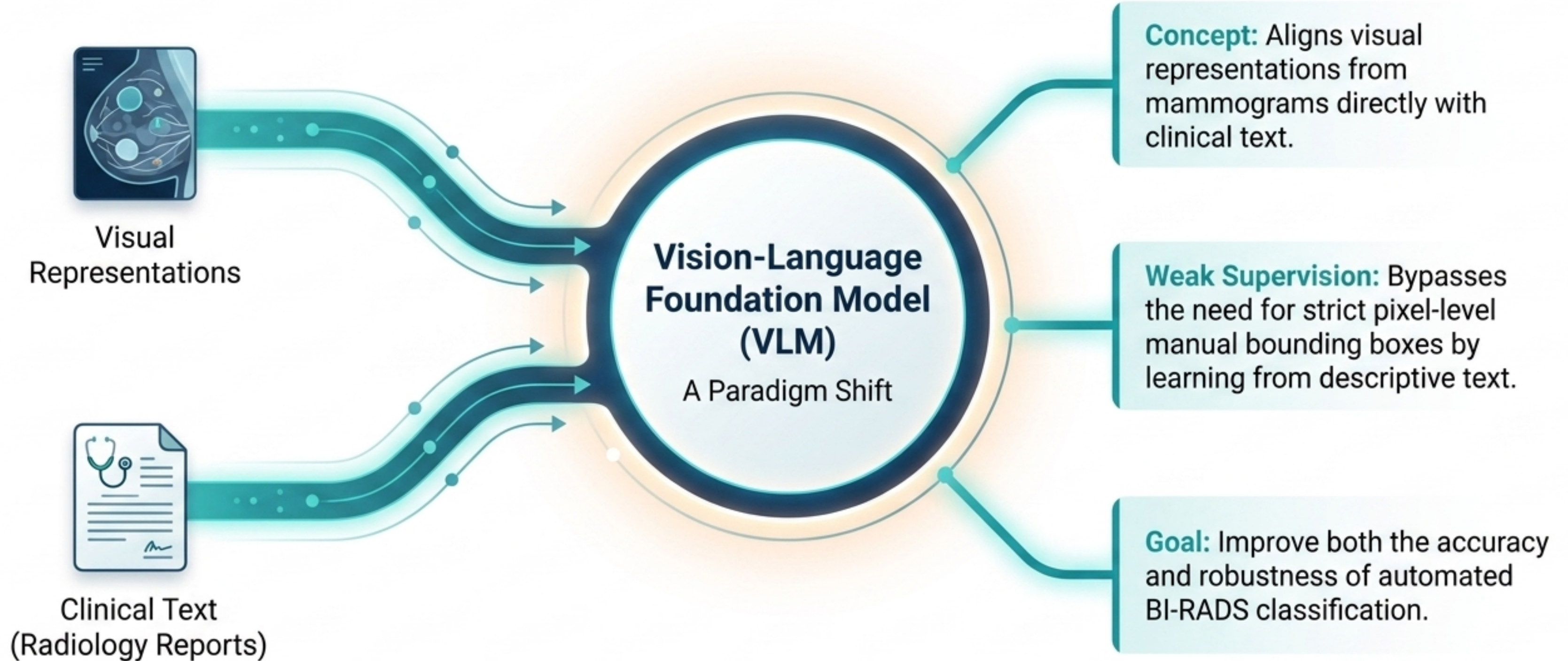


Domain Shift



Traditional models lack robustness, often failing when tested on images from different hospital machines.

Introducing Mammo-CLIP



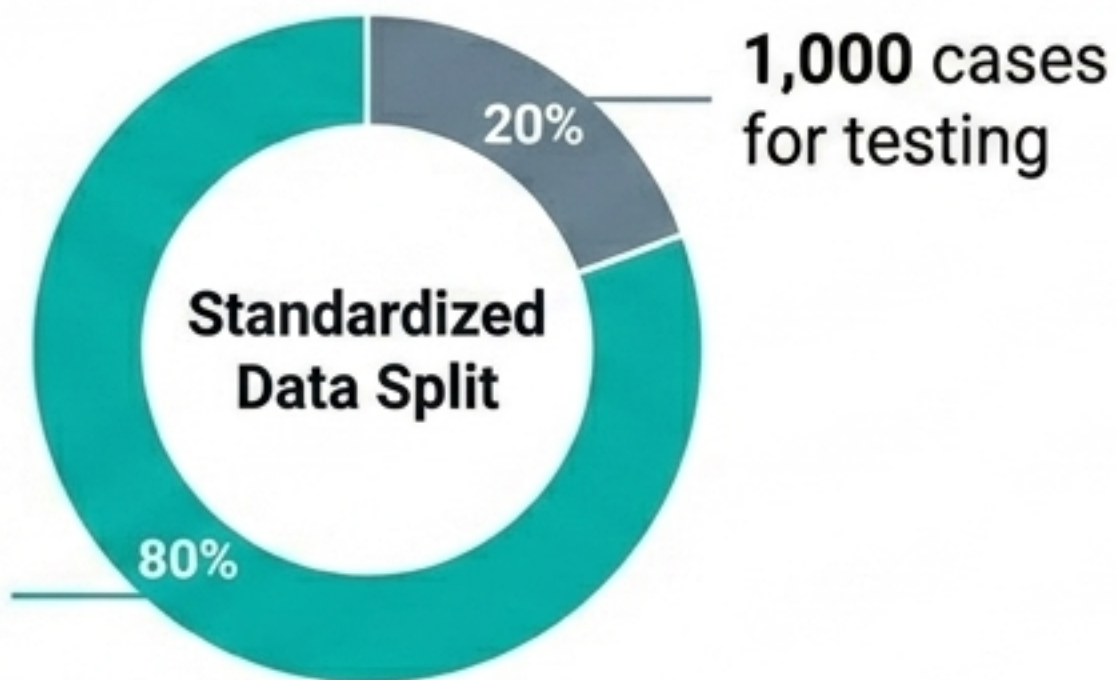
The VinDr-Mammo Dataset

5,000

High-quality Full-Field Digital Mammography (FFDM) cases (approx. 20,000 images).



Annotations: Contains expert-assigned breast-level BI-RADS assessments and finding annotations.



4,000 cases for training

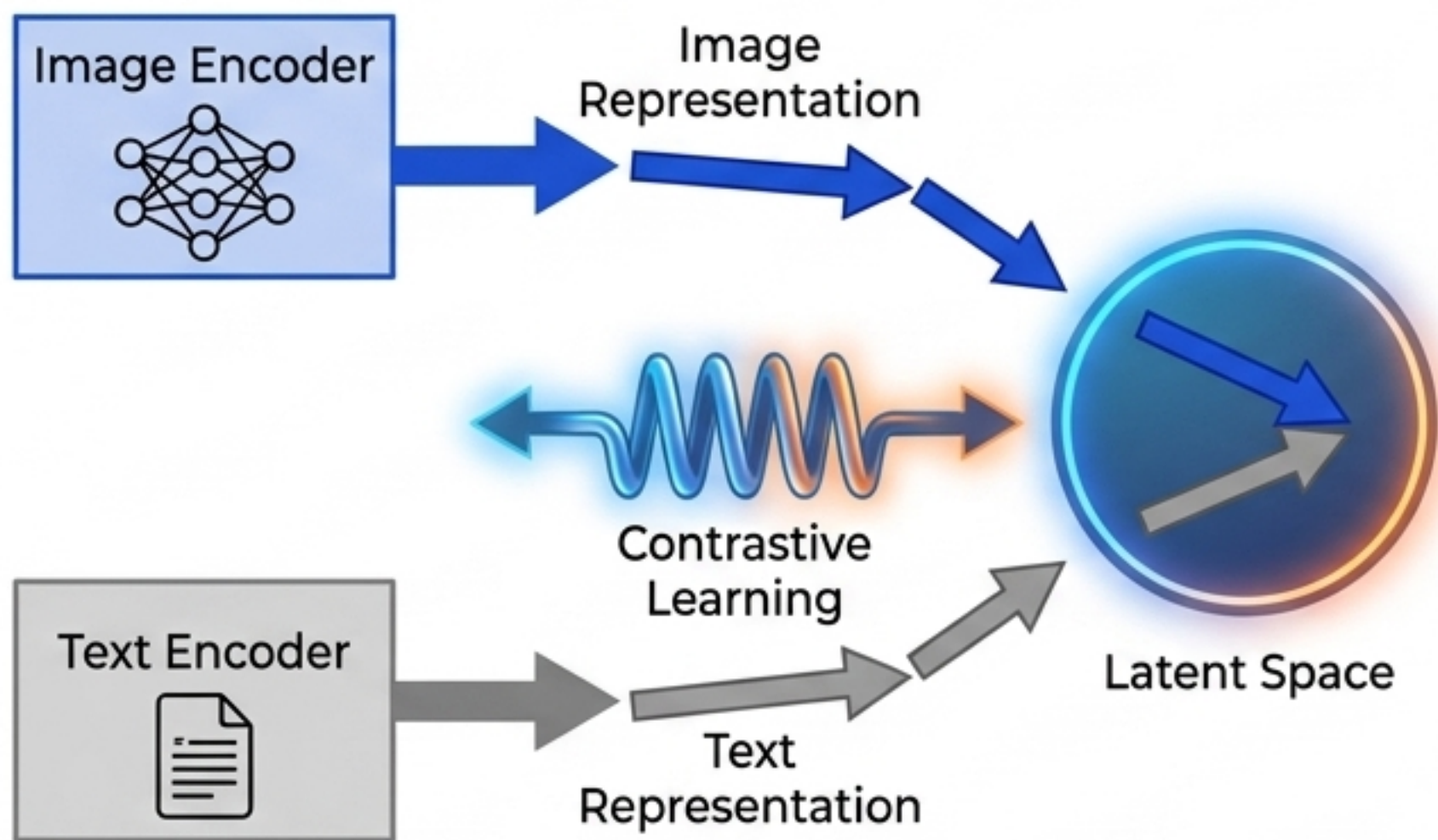
1,000 cases for testing

Diversity: Includes normal, benign, and malignant cases with various breast densities.



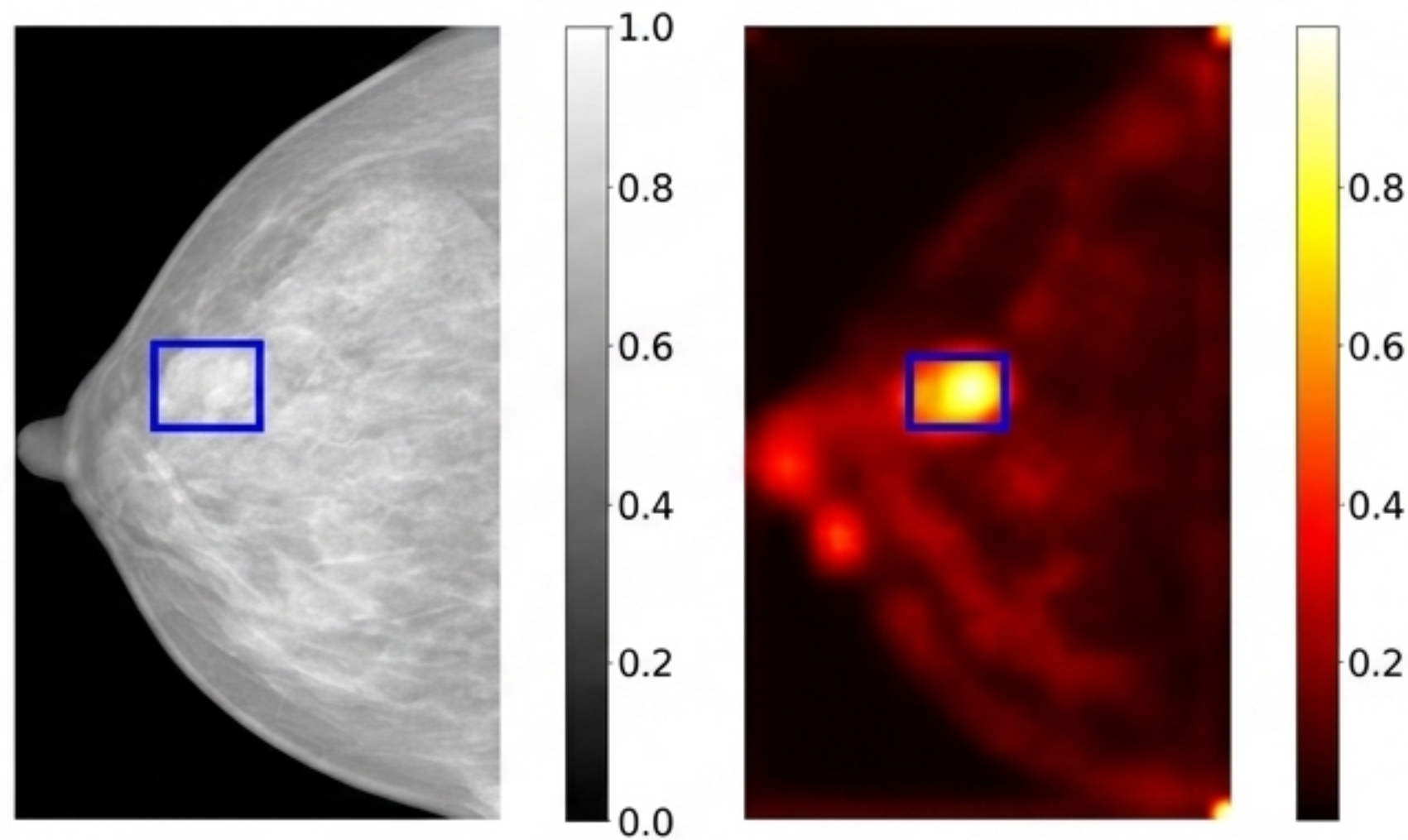
Core Methodology (Part 1): Cross-Modal Alignment

The Architecture



The model **minimizes the distance** between paired images and texts in a **shared latent space**.

The Result



Advantage: Enables the model to understand the **semantics of lesions**, not just pixel patterns.

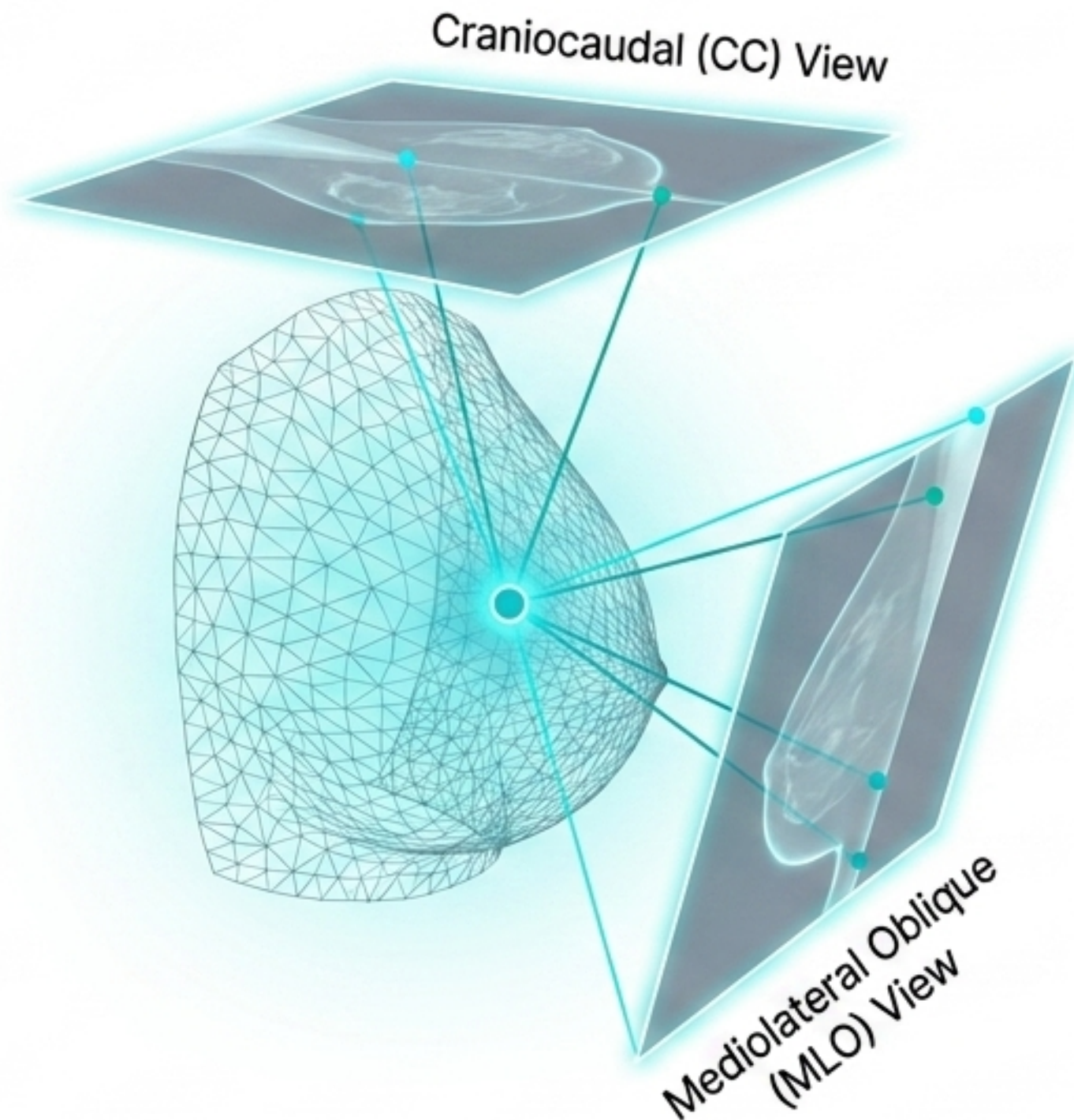
Core Methodology (Part 2): **Multi-View Supervision**

Mimicking Radiologists

Doctors rarely diagnose using a single image. They compare views.

View Fusion

Our framework integrates both Craniocaudal (CC) and Mediolateral Oblique (MLO) views.



Early Feature Fusion

Combines multi-view features early in the network to capture 3D-like spatial relationships.

Benefit

Reduces ambiguity caused by overlapping tissues in a single 2D projection.

Experimental Design

The Tasks

Task 1: BI-RADS Classification
(Predicting Low-risk vs. High-risk profiles based on raw imagery)

Task 2: Lesion Localization

(Accurately finding the coordinates of abnormalities without bounding-box training)

The Competition & Metrics

Baselines Defeated:
Standard CNNs (ResNet-50) and traditional Vision Transformers (ViT).

Evaluation Metrics:

Area Under the Curve (AUC), Accuracy, Sensitivity, Specificity, and mean Average Precision (mAP).

Superior BI-RADS Categorization

0.945

Overall AUC

Mammo-CLIP achieved an AUC of 0.945, outperforming the ResNet-50 baseline (0.862).

91.8%

Accuracy

Accuracy in distinguishing benign from malignant cases.

93.2%

Sensitivity

Crucial improvement for minimizing missed cancer diagnoses.

Conclusion: Multi-modal and multi-view integration significantly boosts classification power.

Robust Localization and Data Efficiency

0.78 mAP

Lesion Localization

Achieved for bounding box predictions, entirely via weak supervision.

25% Data

Data Efficiency

Mammo-CLIP reached baseline performance using only 25% of the labeled training data.

3% Drop

Domain Robustness

Performance dropped by only 3% when tested on an external dataset, compared to a 15% drop in standard CNNs.

Significance in Healthcare



Reduced Workload

Acts as an efficient triage system, allowing radiologists to prioritize high-risk BI-RADS cases.



Cost-Effective AI

Drastically reduces the time and money required to annotate medical datasets.



Standardized Care

Helps minimize human error and inter-reader variability, especially in resource-constrained clinics.



Interpretability

Text-image alignment provides explainable insights into why the AI made a decision.

Next Steps and Future Work

Multi-Institution Validation

Testing the framework on larger, **multi-vendor datasets** (e.g., different mammography machines).

Report Generation

Expanding the model to automatically **generate full narrative radiology reports**.

Multimodal Integration

Combining mammography data with patient **clinical history** (EHR) and **ultrasound imaging** for a **holistic diagnosis**.

Thank You!

Summary: Mammo-CLIP on VinDr-Mammo establishes a robust, data-efficient pipeline for BI-RADS classification.